

CST2213: Business Intelligence Programming 2

Project Phase 1

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1. Project Goals and Scope

1.1 Project Statement

According to multiple reports, Ottawa is considered one of the safest cities to reside/live in 2026. Regardless of this fact, Ottawa has seen a fair share of criminal and safety incidents. For anyone looking to settle in the country's capital, this project will serve as guide to answer 'Which Ottawa neighbourhood is safe to settle in?'

This project applies unsupervised machine learning to eight years of Ottawa Police Service crime data to segment all of Ottawa's neighbourhoods into distinct safety profiles. Using K-Means clustering across 19 crime dimensions and approximately 348,000 recorded incidents from 2018 to 2025, the analysis aims to surface safety divides within the city and quantify where violent and property crime concentrates.

1.2 Objectives

The objective of this project is to use machine learning to group Ottawa neighborhoods based on crime and safety indicators to enable decision making for new and current residents and city officials.

The project will achieve the following objectives:

- Segment all 104 Ottawa neighbourhoods into distinct safety clusters using K-Means clustering applied to 19 crime-type features derived from OPS open data (2018–2025).
- Characterize each cluster's crime composition, distinguishing violent, property, and other crime.
- Assess cluster quality and boundary cases using silhouette scoring to determine how confidently each neighbourhood is assigned to its cluster.
- Deliver an interactive web application allowing exploration of the clustering results at the neighbourhood level.

2. Data Requirements

2.1 Data Sources

The data used for this project will be sourced from the Ottawa Police Service and the City of Ottawa with data ranging from 2017 to 2018. This data will be downloaded as csv and geojson files to enable analysis, model building and geo plotting of these neighbourhoods by cluster.

Data Source/Dataset	Relevance to Project
Criminal Offences	A Criminal Offence is a violation against the Criminal Code of Canada or other federal statute, provincial act, or municipal by-law.
Homicide	A person commits homicide, when directly or indirectly, by any means, causes the death of a human being. This includes 1st-degree murder, 2nd-degree murder, manslaughter and infanticide.
Shootings	A shooting refers to a criminal incident where the victim(s) or property was reported to have been shot or shot at with a firearm.
Auto Theft	Theft of Motor Vehicle refers to the act of taking or another person's vehicle (not including attempts).
Bike Theft	Bike Theft is the act of taking another person's bicycle. Bike Theft figures represent the number of bike stolen.
Hate Crimes	Hate Crime is a criminal violation motivated by hate, based on race, national or ethnic origin, language, colour, religion, sex, age, mental or physical disability, sexual orientation or gender identity or expression, or any other similar factor.
Calls for Service	This dataset contains Calls for Service (CFS) that require a mobile (or dispatched) police response from the Ottawa Police Service (OPS).
Neighbourhood Boundaries	ONS neighbourhood boundaries were derived based on census tracts, physical and demographic similarities, physical barriers

3. Data Collection and Data Sources

3.1 Data Collection Strategy

Public Datasets	Ottawa Open Data Portal (open.ottawa.ca): OPS crime incident dataset (2018–2025) and OPS critical call records. City of Ottawa neighbourhood boundary GeoJSON. Statistics Canada Census 2021 neighbourhood population data.
APIs	No API ingestion required. All datasets are available as direct CSV or GeoJSON downloads from Ottawa Open Data and Statistics Canada. No authentication or API keys are needed.
Access Method	Direct file download from Ottawa Open Data Portal and Statistics Canada website. Files stored locally and loaded into Python (Pandas) as DataFrames for processing. GeoJSON loaded via GeoPandas for spatial operations.

3.2 Data Cleaning Steps

The following cleaning steps are required prior to modelling:

- Neighbourhood name standardization: reconcile naming inconsistencies between the crime dataset, boundary GeoJSON, and population table to enable consistent joins across all three sources.
- Greenbelt and non-residential exclusion: remove Greenbelt-designated neighbourhoods and non-residential parcels (cemeteries, industrial zones) identified as outliers that would distort cluster assignment.
- Pivot and aggregate crime records: transform the raw incident-level records into a feature matrix by pivoting on neighbourhood name and crime type, summing incident counts across the full 2018–2025 window.
- Missing value handling: identify any neighbourhoods with zero counts across all features and assess whether imputation or exclusion is appropriate.
- Feature scaling / normalization: apply StandardScaler (z-score normalization) to the 19-feature matrix before K-Means fitting to prevent high-volume crime types.
- Per-capita ratio computation: join population data to the aggregated crime table and compute crimes-per-capita and critical-calls-per-capita for the police resource allocation analysis, handling any neighbourhoods with missing or zero population counts.
- Data type validation: ensure all crime count columns are cast to integer or float types.

4. Data Exploration and Preliminary Analysis

4.1 Key Trends and Insights

Initial exploration of the dataset revealed the following key observations:

- Extreme right-skew in incident distribution: Centretown alone recorded 52,597 incidents. The distribution of total incidents per neighbourhood is severely right-skewed, confirming the need for StandardScaler normalization before model fitting.
- Distinct urban-suburban-rural safety gradient: a clear geographic pattern emerges in the data. The urban core (Centretown, Byward Market, Lowertown) shows the highest incident densities, the suburban ring (Barrhaven, Stittsville, Orleans) sits in the mid-range, and the outermost rural areas (Kinburn, Sarsfield, Pierces Corners) record the fewest incidents.
- Property crime dominates citywide: across all neighbourhoods, property crime (theft, break and enter, auto theft, bike theft, mischief, arson) accounts for the majority of incidents.
- Optimal cluster count identified at k=6: the Elbow method applied to within-cluster sum of squares (WCSS) and silhouette analysis indicates that six clusters best balance separation and cohesion.
- Auto theft concentration: New Barrhaven/Stonebridge alone recorded 605 auto thefts over the study period, more than double any other neighbourhood.

4.2 Data Issues Identified and Handling Strategy

- Extreme outlier (Centretown): Retained in model with StandardScaler normalization; documented as a statistical cluster of one
- Non-residential / Greenbelt parcels: Explicitly excluded from the K-Means feature matrix before model fitting
- Neighbourhood name inconsistencies across datasets: Manual reconciliation using a name-mapping lookup table; standardize on OPS spelling as primary key
- Per-capita distortion in low-population areas: Flag affected neighbourhoods (non-residential areas with populations near zero) and exclude from per-capita analysis

4.3 Planned Visualizations

The following visualizations will be produced during the analysis and included in the final deliverable:

- Choropleth map: each of the Ottawa neighbourhoods shaded on a six-step green-to-dark-maroon gradient representing its safety cluster assignment.
- Stacked bar chart: total crime composition per cluster broken down by category (violent, property, other), sorted from safest to highest-crime cluster. Reveals how crime type mix changes across the safety gradient.
- Scatter plot with OLS trend line: crimes per capita (x-axis) vs. critical police calls per capita (y-axis), one point per cluster. Identifies clusters above or below expected policing levels relative to their crime volume.
- Z-score feature heatmap: Exposes the distinct crime fingerprint of each cluster.
- Silhouette score bar chart: one bar per neighbourhood grouped by cluster, sorted from lowest to highest score, with red bars marking negative scores. Quantifies assignment confidence and surfaces boundary-case neighbourhoods.
- Interactive Streamlit web application: embeds the choropleth map with neighbourhood-level drill-down, cluster summary cards, and filterable crime-type breakdowns for public and stakeholder exploration.

5. Tools and Technologies

5.1 Technology Stack

Category	Tool / Library	Purpose in This Project
Programming Language	Python	Core language for all data ingestion, cleaning, modelling, and visualization tasks
Data Manipulation	Pandas, NumPy	Loading and pivoting OPS crime records into the 104 × 19 feature matrix; joining population and police call tables; per-capita calculations

Machine Learning	Scikit-learn (KMeans, StandardScaler, silhouette_score)	Feature normalization, K-Means model fitting, optimal k selection (Elbow + silhouette), and cluster quality evaluation
Data Visualization	Matplotlib, Seaborn	Stacked bar charts, z-score heatmap, scatter plot with OLS trend line, silhouette score bar chart
Geospatial Mapping	Folium, GeoPandas, Plotly	Loading neighbourhood boundary GeoJSON; rendering choropleth safety cluster map for web app and report
Web Application	Streamlit	Interactive web app for neighbourhood-level cluster exploration
IDE / Notebooks	Jupyter Notebook, VS Code	Exploratory data analysis, model development, and documentation